Controlled Packet Capture and 802.11 Protocol Analysis Lab - Understanding 802.11 Frames

Understanding the fundamental structure of a 802.11 frame is essential to accurately interpret a packet capture. In this post we will explore the structure of a MAC frame and frame control field to highlight thier importance. In subsequent posts we will discuss in further detail each field via packet capture analysis.

802.11 MAC Frame

What exactly is an 802.11 MAC frame?

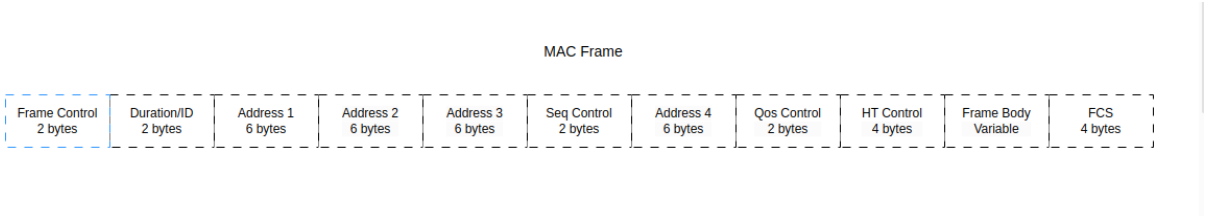
Think of a 802.11 MAC frame as a package being sent between devices over a Wi-Fi network. Just like a shipping package has information about where it's coming from, where it's going, and what's inside, a MAC frame organizes the information needed to communicate between wireless devices.

When devices communicate over a wireless netowork they exchange these frames. Some are used to discover nearby networks, others help devices connect and communicate, and some carry the actual data being sent.

What does its structure look like? Let's have a look at the components that make up a frame.

It is important to note that not every frame looks exactly the same. Its structure can change depending on what the device is trying to accomplish.

Basic Structure of MAC Frame



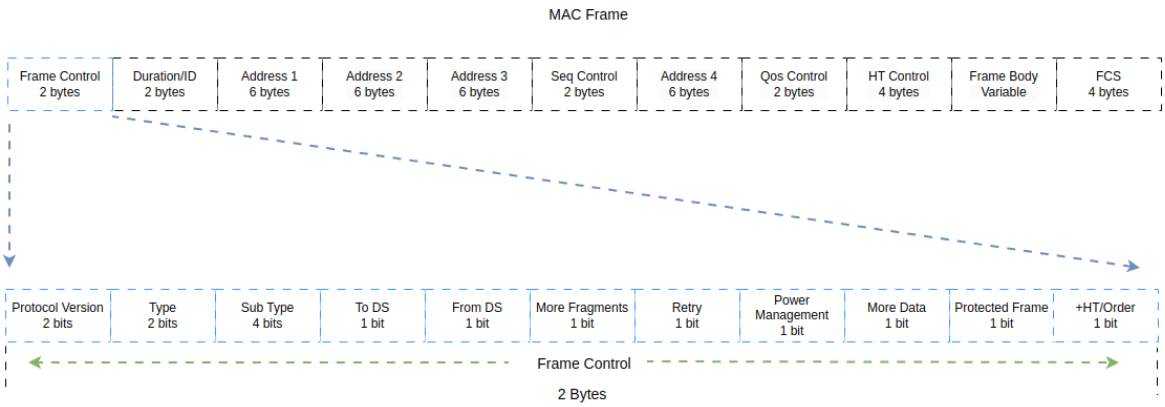
The goal of this lab is to become familiar with these frames, understand how they are put together, and learn how to recognize them when looking at a wireless packet capture in Wireshark.

Frame Control

Let's jump right in! Starting with Frame Control.

What is the purpose of this 2 byte field?

To simplify, frame control set's the stage! It tells us what kind of frame we're looking at and what it's intended to do.



Frame Control Field Breakdown

The Frame Control field is the first 2 bytes of an 802.11 MAC frame. It tells us what type of frame we're looking at and provides additional information about how that frame is being handled.

- **Protocol Version (2 bits)** - Version of the 802.11 protocol being used.
- **Type (2 bits)** - Category of the frame: Management, Control, or Data.
- **Subtype (4 bits)** - Provides more detail about the specific purpose of the frame, such as Beacon, Probe Request, ACK, or Data.
- **To DS (1 bit)** - Direction of the frame. Is the frame being sent **toward** the Distribution System.
- **From DS (1 bit)** - Direction of the frame. Is the frame coming **from** the Distribution System.
- **More Fragments (1 bit)** - Are additional fragments of the same frame expected.
- **Retry (1 bit)** - Is the frame being retransmitted.
- **Power Management (1 bit)** - Is the sending device entering or operating in a power-saving mode.
- **More Data (1 bit)** - Additional buffered data is waiting for the receiving device.
- **Protected Frame (1 bit)** - Is the frame body is protected using encryption or other security mechanisms.
- **Order (1 bit)** - Special ordering requirements for certain frame types.

When examining packets in wireshark we will start with the Type and Subtype to identify the type of activity we are looking at. Both fields are important and warrant further discussion.

Frame Type and Subtype

Type and Subtype fields work together to identify what kind of frame we're looking at and what that frame is being used for.

Type (2 bits)

Type will tell us the general category of the frame. There are three primary frame types.

- **Management (00)** -Discover networks, establish connections, and manage wireless communication.
- **Control (01)** - Coordinate wireless communication and manage access to the airwaves.
- **Data (10)** - Carry information between wireless devices.

Subtype (4 bits)

Subtype provides more detail about the specific purpose of the frame within its Type category.

Management Frame Subtypes

- Association Request - A device requests to join a wireless network.
- Association Response - The AP responds to a device's connection request.
- Probe Request - A device searches for nearby wireless networks.
- Probe Response -An AP responds to a network discovery request.
- Beacon - An AP periodically advertises its wireless network.
- Disassociation - Terminates an existing wireless association.
- Authentication - Used during the authentication process.
- Deauthentication - Ends an authentication relationship.

Control Frame Subtypes

- Request to Send (RTS) - Requests permission to use the wireless medium.
- Clear to Send (CTS) - Indicates that the wireless medium is available for the requested transmission.
- Acknowledgment (ACK) - Confirms successful frame reception.
- Block ACK - Acknowledges multiple frames.

Data Frame Subtypes

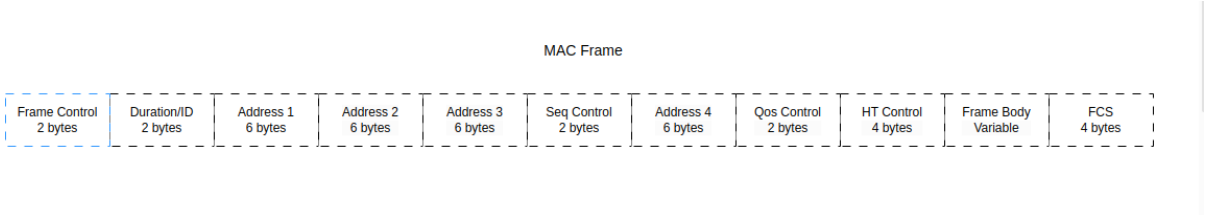
- Data - Carries user data.
- Null Data - Contains no user payload but can be used for signaling.
- QoS Data - Carries data with Quality of Service information.
- QoS Null - Contains no user payload but includes QoS information.

Think of (Type) as the general category and (Subtype) as the specific action. For example, a Beacon frame has a (Type of Management) and a (Subtype of Beacon). This understanding is especially useful when filtering and analyzing wireless traffic in wireshark.

MAC Frame

Now that we have a better unerstanding of the Frame Control Field, let's have a look at the remaining MAC Frame.

MAC Frame



Remaining 802.11 MAC Frame Fields

Beyond the Frame Control field, 802.11 MAC frames contain several other fields that help identify devices, manage communication, and carry information across the wireless network.

- **Duration/ID (2 bytes)** - Indicates how long the wireless medium is expected to remain occupied. Other devices use this information to determine when the channel may become available.
- **Address 1 (6 bytes)** - Typically identifies the receiving device or destination of the frame.
- **Address 2 (6 bytes)** - Typically identifies the transmitting device or source of the frame.
- **Address 3 (6 bytes)** - Provides additional addressing information, often identifying the source, destination, or wireless access point (BSSID), depending on the frame.
- **Sequence Control (2 bytes)** - Helps identify frames and keep track of their order. It also assists with handling fragmented frames and retransmissions.
- **Address 4 (6 bytes, optional)** - Used in certain wireless configurations when additional address information is needed.
- **Frame Body (variable)** - Contains the ctual information being transmitted. Depending on the frame type, this could include network information, connection details, or user data.
- **Frame Check Sequence (FCS) (4 bytes)** - Used to detect errors or corruption in a frame during transmission.

Important things to remember

- Not every frame contains all four address fields.
- Meaning of the MAC address fields can change depending on the To DS and From DS flags.
- Frame Body varies depending on whether the frame is Management, Control, or Data.

When examining captures in wireshark it is important to pay attention to how MAC addresses relate to the transmitting device, receiving device, and BSSID. This will help with understanding who is communicating with whom over the wireless network.

Conclusion

We have explored the overall structure of the MAC frame, examined its fields, and highlighted the importance of the Frame Control field. In subsequent posts, we will perform packet captures, review them using wireshark, and dive deeper into the details of the various MAC frame fields.